TEXTBOOK OF AGRICULTURAL STATISTICS · CHAPTER 18

5-MINUTE REVISION

Analysis of Variance

From the chapter's own revision points, formulae and questions.

BOOK-ONLY EDITION · Verified against the textbook

Selected from Chapter 18 · Chapter Summary in the textbook.

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STAT 3202 · Chapter revision

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Quick revision · 1

- ANOVA compares population means by partitioning the total variation.
- The one-way null hypothesis is $H_0 : \mu_1 = \cdots = \mu_v$.
- A mean square is a sum of squares divided by its degrees of freedom.
- The treatment F-ratio compares treatment mean square with error mean square.
- The usual exact F test assumes independent normal errors with a common variance under the specified model.

Quick revision · 2

- A significant overall F indicates that at least one pair of means differs; it does not identify the pairs.
- The additive two-way model separates treatment and block variation.
- With one observation per combination, interaction cannot be estimated separately from error.
- Check the design and residuals before interpreting the F tests.

Important formulae

For a one-way classification with v treatments and n observations, let T_i be the total and n_i the number of observations in treatment i, and G the grand total.

$$CF = \frac{G^2}{n}, \qquad SS_{Total} = \sum_{i,j} y_{ij}^2 - CF$$

$$SS_{Treatment} = \sum_{i=1}^{v} \frac{T_i^2}{n_i} - CF, \qquad SS_E = SS_{Total} - SS_{Treatment}$$

$$MS_{Treatment} = \frac{SS_{Treatment}}{v-1}, \qquad MSE = \frac{SS_E}{n-v}, \qquad F = \frac{MS_{Treatment}}{MSE}$$

For a balanced additive two-way classification with v treatments and r blocks, one observation per combination, and block totals B_j:

$$SS_{Blocks} = \sum_{j=1}^{r} \frac{B_j^2}{v} - CF$$

$$SS_E = SS_{Total} - SS_{Treatment} - SS_{Blocks}, \qquad edf = (v-1)(r-1)$$

Self-check from the chapter

1. ANOVA stands for Analysis of _____. **[Variance]**
2. The null hypothesis in a one-way fixed-effects ANOVA states that all population means are _____. **[Equal]**
3. A sum of squares divided by its degrees of freedom is called a mean _____. **[Square]**
4. The denominator of the usual treatment F-ratio is the _____ mean square. **[Error]**